

LIMITATIONS OF THE PROJECT

1. No conversation memory
2. Fixed chunk sizes
3. PDF parsing limitations
4. Poor handling of very broad questions
5. No user feedback loop
6. No streaming responses

SOLUTIONS

1. No conversation memory
 - Include history-aware retriever
`create_history_aware_retriever()`
 - This rewrites the user's follow-up question into a standalone question before retrieval.
2. Fixed chunk sizes
 - Different documents need different chunking.
 - Try average size
`Chunk_size=700`
`Chunk_overlap=150`
3. PDF parsing limitations
 - Some PDFs contain Tables, Images, Diagrams, Multi-column layouts and PyPDFLoader often extracts them poorly.
 - Consider OCR for scanned PDFs, better document parsers and table extraction tools
4. Poor handling of very broad questions
 - Retriever returns only a few chunks and large portions of the PDF are ignored.
 - Adaptive retrieval which retrieves many sections, combines and generates structured answer

5. No user feedback loop

- System never learns which answers were useful
- Add helpful and not helpful to store feedback for future evaluation

6. No streaming responses

- User waits then receives complete answer
- Use streaming with Groq to see answer token by token